Neural Wave AI Hackathon - Lifeware Challenge

Surprise challenge t7: The Tax Audit



Lifeware Company Logo

1 Breaking News

Message from the CFO of NeuralWave Life:

The tax authorities of Switzerland, Italy and Germany have announced an audit. For every payout since 1994 they want to know what we withheld and when we paid it to them, and for every policy and year what we reported. Neuron's tax module is the only system that knows, and it will be switched off together with Neuron. Please recalculate exactly what Neuron did, and not what the law says today. Due to privacy concerns of the sensible tax data, this **must** be done with custom models or open weights models.

NeuralWave Life sells its policies not only in Ticino. Its customers live in Switzerland, Italy, Germany and Luxembourg, and some of them moved from one country to another while their policy was running. Every country taxes a payout of a life insurance in its own way, and the rules changed over the years. Neuron had a tax module that handled all of this, and like the rest of Neuron, nobody documented how it calculates.

2 Why This Matters

A tax that was withheld wrongly is not a rounding difference: it is money that belongs to the customer or to the state. In a real migration the tax module is often the part with the most history, because every change of the law was patched into the old system, sometimes late, sometimes wrongly, and sometimes corrected afterwards. The new system has to show the auditor exactly what the old one did, including its mistakes.

3 The Data

The tables of the tax module have been in your extract from the very beginning. You may already have seen them:

`NRHOLD`	the policyholder of a policy, with its history.
`NRADDR`	the residence of every person, with its history: country, region, postal code, tax reporting status and church membership
`NRTAXR`	the tax rates per country and tax type, with validity periods and versions
`NRTAXB`	the tax bookings of Neuron. As in `NRBOOK`, the amounts are missing.

4 The Task

t7 Taxes - (Surprise challenge #1)

Goal: for every payout calculate the taxes Neuron withheld and the day it paid them to the tax office, and for every policy and calendar year the tax amounts Neuron withheld, accrued or reported.

Why: the audit.

Result: two functions.

- `t7_tax_payout`: {`(policy_id, event_nr)`: `(country, taxable_base, withholding, surcharges, stamp_duty, remittance_date)`}, file `t7_tax_payout.csv`. `country` is the country of residence Neuron used, `remittance_date` is `None` if nothing was withheld.
  Example: `BL10000086,121,DE,5628.16,1407.04,77.38,0.00,2006-08-10`.
- `t7_tax_annual`: {`(policy_id, year)`: `(ch_stp, ch_vst, it_sos, it_bol, de_kst, de_sol, de_kis)`}, file `t7_tax_annual.csv`, only years with at least one amount.
  Example: `BL10000044,2019,0.00,0.00,0.00,616.11,0.00,0.00,0.00`.

5 Submission

Add two functions to your `submission.py`:

```
def t7_tax_payout(legacy_dir) -> dict[tuple[str, int], tuple]

def t7_tax_annual(legacy_dir) -> dict[tuple[str, int], tuple[Decimal, ...]]
```

The same rules apply as for the other tasks: everything must be calculated from `legacy_dir`, no target values in the code, and the tools you build count for t6.

6 Evaluation

A row counts only if every column is exact: the country, the amounts to the cent and the remittance date. t7 is scored on the public and private policies and reported separately from the other tasks.

7 Changes to the leaderboard

Since this is a surprise audit, we do not have all target data. Instead, whenever we get a new implementation, it has to be sent to an actuary of NeuralWave Life who computes the values by hand. To simulate this, the number of targets is drastically reduced to only a handful of targets, which can change between evaluations, because the actuary will want to know if different patterns work correctly. The time between evaluations is increased to two hours. Since one person cannot evaluate hundreds or thousands of policies in a short period of time and the response time is not instantaneous, when communicating over email.